

THE SUN

The Sun is the hottest and largest object in our Solar System. Its fiery surface bathes the planets around it in light, and its gravity shapes their orbits. The Sun is now about halfway through its life. In about 5 billion years it will turn into a red giant, before puffing its outer layers into space, leaving behind only a ghostly cloud called a planetary nebula.

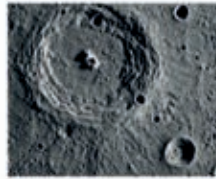
**THE SUN IS SO HUGE
THAT EARTH COULD
FIT INSIDE IT ONE
MILLION TIMES**

The planets

Around 4.6 billion years ago, a great cloud of dust and gas formed into the Sun. The parts that were not used began to form into clumps, which grew into planets orbiting the Sun. The four planets closest to the Sun formed from rock and metal. The four bigger outer planets formed from gases.

MERCURY

Mercury is the nearest planet to the Sun and the smallest in the Solar System – it is about as wide as the Atlantic Ocean. Mercury is a rocky world that has no atmosphere or water.



BRAHMS CRATER
Mercury is covered in craters made by debris crashing into its surface.

MERCURY

ROCKY PLANET

DISTANCE FROM THE SUN: 69.8 million km
(43.3 million miles)

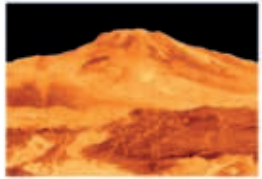
DIAMETER: 4,879 km
(3,030 miles)

TIME TAKEN TO ORBIT THE SUN: 87.97 Earth days

NUMBER OF MOONS: 0

VENUS

Venus is the second planet from the Sun. It is about the same size as Earth and made from similar materials, but its atmosphere is made of carbon dioxide – the gas that we breathe out.



MAAT MONS
Venus has more than 1,600 volcanoes, the highest of which is Maat Mons.



VENUS

ROCKY PLANET

DISTANCE FROM THE SUN: 108.9 million km
(67.6 million miles)

DIAMETER: 12,104 km
(7,520 miles)

TIME TAKEN TO ORBIT THE SUN: 224.7 Earth days

NUMBER OF MOONS: 0

JUPITER

Jupiter is the largest planet in the Solar System – it could hold around 1,300 Earths. It is a giant ball of gas ringed by colourful bands of chemical gases that race around it as fierce winds.



GREAT RED SPOT
This is a giant storm several times bigger than Earth, which has been raging for 300 years.

JUPITER

GAS GIANT

DISTANCE FROM THE SUN: 816 million km (507 million miles)

DIAMETER: 142,984 km
(88,845 miles)

TIME TAKEN TO ORBIT THE SUN: 11.86 Earth years

NUMBER OF MOONS: 67+

SATURN

The second-largest planet in the Solar System, Saturn is not dense – it would float in a planetary-sized bathtub. It is surrounded by a system of rings that extend thousands of kilometres from the planet but are only 9 m (30 ft) thick.



RINGS OF SATURN
Saturn's rings are made up of ice crystals and rock.



SATURN

GAS GIANT

DISTANCE FROM THE SUN: 1.5 billion km
(932 million miles)

DIAMETER: 120,536 km
(74,900 miles)

TIME TAKEN TO ORBIT THE SUN: 29.46 Earth years

NUMBER OF MOONS: 62+

DISTANCE FROM THE SUN

The distances between the planets are huge, becoming bigger as we move out through the Solar System. If the Sun were the size of a grapefruit, Neptune would be 14.5 km (9 miles) away.

